4 **(a)** A sample of sand has a volume of $0.050\,m^3$. The density of the sand is $1900\,kg/m^3$. The specific heat capacity of the sand is $1500\,J/(kg\,°C)$.

(i) Calculate the mass of the sample of sand.

mass = ... [2]

(ii) Calculate the thermal capacity of the sample of sand.

thermal capacity = ... [2]

(iii) The initial temperature of the sample of sand is 7.0 °C. The sample of sand is heated using an electrical heater. The power of the heating element is 50 W.

Calculate the time taken to increase the temperature of the sand to 19.0 °C.

time = ... [3]

(b) In some countries, the soil is too cold for plants to grow well. In these countries, plants are grown in plastic pots and kept inside. The pots, containing soil, are placed on sand. The sand is heated using an electrical heater, as shown in Fig. 4.1.

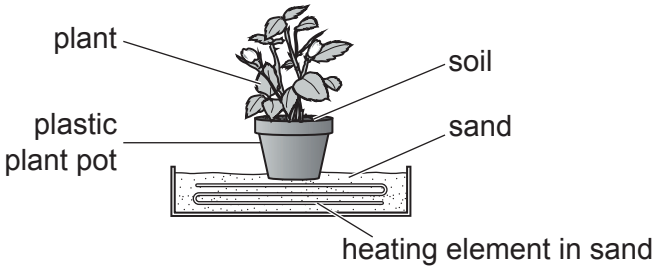


Fig. 4.1

(i) Describe, in terms of molecules, how thermal energy is transferred from the heated sand through the base of the plastic pot.

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..... [2]

(ii) The heating element in Fig. 4.1 remains switched on. The temperature of the sand remains constant at a value above room temperature.

Explain why the temperature of the sand remains constant.

.....

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..... [2]

[Total: 11]